

4/9/2025

Network Design for Food Chain Headquarters

Student Name: Samarth Gunvantbhai Desai

SRN: 21079043

Module Leader: Dr. Seyedali Pourmoafi

Module Name: Network Protocols and Architectures

Module Code: 6COM1067

Table of Contents

INTRODUCTION	2
CHAPTER 1: LOCAL AREA NETWORK (LAN)	3
1.1 DESIGN DESCRIPTION AND JUSTIFICATION	3
1.2 SCALABILITY CONSIDERATIONS	4
1.3 BENEFITS OF HIERARCHICAL DESIGN	5
1.4 NETWORK HARDWARE SELECTION AND EVALUATION	7
CHAPTER 2: WIDE AREA NETWORK (WAN).....	8
2.1 DESIGN DESCRIPTION AND JUSTIFICATION	8
2.2 NETWORK HARDWARE SELECTION AND EVALUATION	9
CONCLUSION	10
REFERENCES	11

Table of Figures

Figure 1: Logical LAN Diagram	3
Figure 2: Physical LAN Diagram	3
Figure 3: Logical WAN Diagram	8

Introduction

This report presents the design and justification of a scalable, redundant, and high-performance network infrastructure for the headquarters of a large food chain enterprise. The organization oversees critical operations such as inventory management, sales, logistics, and customer services, requiring a robust and fault-tolerant communication system. The proposed solution includes a hierarchical Local Area Network (LAN) architecture, and a Wide Area Network (WAN) interconnection based on Dense Wavelength Division Multiplexing (DWDM) to link the headquarters with the main premises.

The network design aligns with enterprise best practices and leverages industry-standard hardware such as Cisco Nexus, Catalyst, and ASR series devices, as well as Palo Alto firewalls and Ciena optical systems. It incorporates both logical and physical network diagrams, supported by detailed hardware specifications and cabling standards. The report addresses key requirements such as redundancy, scalability, and security, ensuring the infrastructure can support current demands while remaining adaptable to future growth.

Chapter 1: Local Area Network (LAN)

Logical and Physical Diagrams

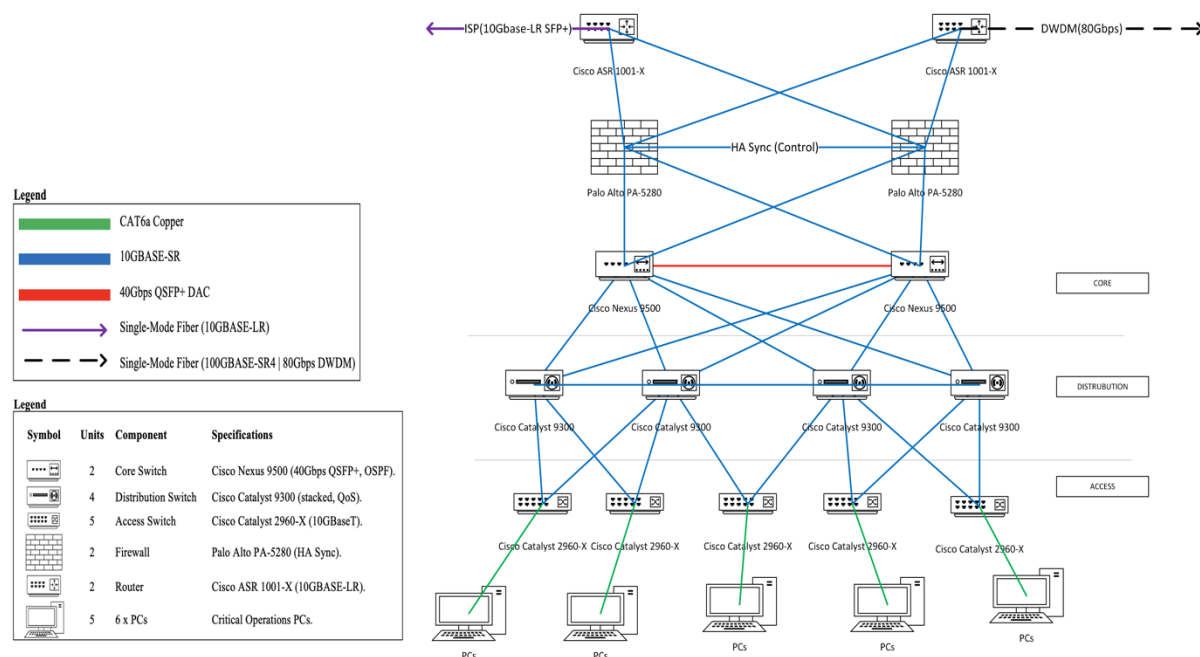


Figure 1: Logical LAN Diagram

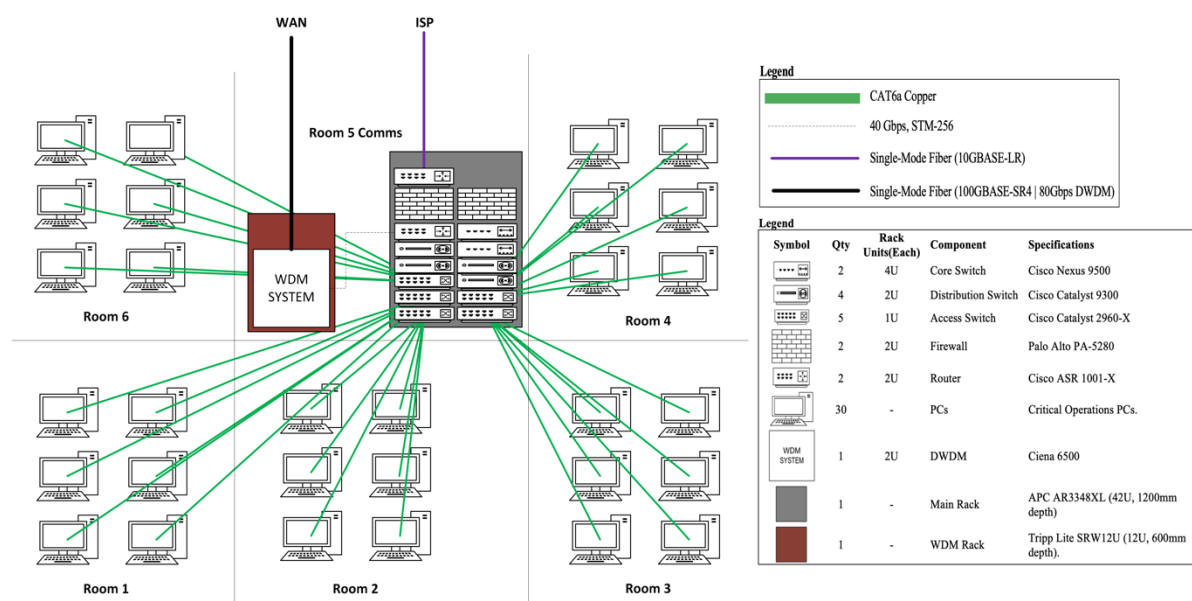


Figure 2: Physical LAN Diagram

1.1 Design Description and Justification

The proposed LAN infrastructure for the headquarters of the food chain adheres to the three-tier hierarchical network model, incorporating core, distribution, and access layers. This modular

design enhances redundancy, scalability, and ease of management—key requirements highlighted in the assignment specifications.

Redundancy is achieved through dual connections across all layers. Core switches (Cisco Nexus 9500) are configured in a fully meshed topology with distribution switches (Cisco Catalyst 9300), ensuring that no single point of failure exists. Similarly, distribution switches connect redundantly to multiple access switches (Catalyst 2960-X), which provide network access to critical devices in each office, such as six desktop PCs per room.

Room 5 is designated as the communications (comms) room, housing all core infrastructure components including firewalls, routers, and switches, mounted in a secure APC AR3348XL 42U rack. The WAN uplink via WDM and ISP is integrated into this room for efficient data routing and centralized control.

- All cabling adheres to standards defined in the legend:
- CAT6a for short-range copper connections
- 10GBASE-SR is used for short-range multimode fibre connections between switches within the same rack or floor. 10GBASE-LR supports long-range single-mode fibre uplinks across different floors or building areas.
- QSFP+ DAC for high-speed inter-switch links

These mediums provide a blend of performance and cost-efficiency and are suitable for high-throughput operations such as real-time inventory and sales processing.

1.2 Scalability Considerations

The LAN architecture has been meticulously designed to ensure both vertical and horizontal scalability, catering to the dynamic operational requirements of the food chain headquarters. Scalability is a core consideration, not only to meet current performance demands but also to accommodate future technological growth, workforce expansion, and increasing data traffic resulting from business growth and digital transformation.

Vertical scalability is achieved through the deployment of modular core and distribution layer devices. The Cisco Nexus 9500 core switches, for instance, support multiple QSFP+ ports with up to 40Gbps bandwidth and modular chassis slots that allow hardware upgrades without system downtime. These switches are fully compatible with higher-speed optics and can be upgraded to support newer standards like 100GBASE-SR4 or 400G modules in the future. Similarly, the Cisco Catalyst 9300 distribution switches feature stacking capabilities, which allow multiple switches to operate as a single logical unit. This stacking flexibility ensures that

new switches can be added seamlessly to support additional floors, departments, or business units without reconfiguring the entire network.

Horizontal scalability is addressed at the access layer using Cisco Catalyst 2960-X access switches, each offering multiple 10GBaseT ports for connecting end-user devices. These switches are strategically distributed across office rooms, each supporting 6 critical PCs, as represented in the logical LAN diagram. As more endpoints are introduced, such as VoIP phones, IoT devices, or thin clients, these access switches provide sufficient port availability and VLAN support for segmentation and security.

The fibre-optic backbone, which includes 10GBASE-SR and 10GBASE-LR connections between core and distribution layers, ensures that the network can handle higher data throughput without encountering congestion. The 40Gbps QSFP+ DAC links at the core layer provide ample headroom for increased inter-switch traffic and can scale to higher bandwidths with minimal physical modification.

Additionally, the infrastructure is rack-optimized, with most networking equipment housed in the central APC AR3348XL 42U rack in Room 5. This design choice centralizes physical infrastructure, allowing for streamlined management and the future installation of blade servers, backup systems, and virtualization platforms such as VMware ESXi or Microsoft Hyper-V. With the inclusion of modular cabling and structured patch panels (as implied by your physical diagram), adding new devices or services can be executed efficiently and without disrupting ongoing operations.

Therefore, the proposed architecture is not only scalable but also resilient, modular, and forward-compatible, offering longevity and adaptability to technological advancements and enterprise needs.

1.3 Benefits of Hierarchical Design

The implementation of a three-tier hierarchical model—comprising the core, distribution, and access layers—offers several operational and strategic advantages, many of which are reflected in the logical and physical LAN diagrams provided.

Performance Optimisation

The separation of network functions across layers ensures that each tier focuses on specific tasks. The core layer, using Cisco Nexus 9500 switches with high-speed QSFP+ DAC links, is optimized for fast and efficient packet forwarding between different segments of the network. The distribution layer, implemented with Catalyst 9300 switches, handles routing, inter-VLAN

communication, and policy enforcement such as QoS for latency-sensitive traffic like VoIP. Meanwhile, the access layer, using Catalyst 2960-X switches, focuses on reliable delivery of services to endpoints (i.e., user PCs), ensuring network efficiency and reduced latency.

Enhanced Troubleshooting and Isolation

One of the defining advantages of a hierarchical topology is fault domain isolation. If an issue arises at the access layer—such as a faulty port on an access switch—it can be contained and addressed without affecting the distribution or core layers. This layered approach significantly reduces mean time to repair (MTTR) and simplifies root cause analysis. Additionally, core layer faults can be bypassed through redundant links, as demonstrated in your logical diagram where dual paths exist between switches.

Security and Control

Placing Palo Alto PA-5280 firewalls between the core switches and border routers allows for deep packet inspection, intrusion prevention, and policy enforcement. These next-generation firewalls are configured in high-availability (HA) mode, which aligns with the assignment's redundancy requirement. The firewalls and routers are strategically positioned to secure both inter-VLAN and WAN-bound traffic, ensuring controlled access to sensitive systems such as inventory management, financial applications, and employee databases.

Ease of Maintenance and Upgradeability

In this architecture, hardware components can be upgraded at the individual layer without affecting the rest of the network. For example, upgrading a distribution switch for higher throughput would not necessitate modifications to the access switches or core infrastructure. This modularity ensures business continuity and lowers operational risk during maintenance windows.

Scalability Integration

The design also benefits from the scalability characteristics inherent to hierarchical networks. As business operations expand, new departments or facilities can be integrated by simply extending the access layer and linking it back to existing distribution and core layers using standardised fibre modules and copper cabling, as outlined in the cable legend.

As Cisco (2023) states, the hierarchical model is foundational in scalable network architecture, promoting resilience, security, and adaptability. This structure has become a standard in enterprise deployments due to its support for predictable performance, efficient troubleshooting, and scalable growth across diverse and high-demand environments.

1.4 Network Hardware Selection and Evaluation

The hardware selected for this LAN design was chosen to ensure high availability, performance, and future scalability, reflecting real-world enterprise standards. At the core layer, the Cisco Nexus 9500 switches provide high-throughput interconnectivity using 40Gbps QSFP+ links. These modular, high-capacity switches are ideal for handling backbone traffic and support routing protocols like OSPF, making them a reliable choice for the network’s central infrastructure (Cisco, 2023).

The distribution layer uses Cisco Catalyst 9300 switches, offering advanced Layer 3 features, QoS support, and stack ability. These switches enable efficient inter-VLAN routing and policy enforcement while allowing for horizontal scalability through physical stacking. Their positioning between access and core layers improves both redundancy and performance.

For end-device connectivity, Cisco Catalyst 2960-X switches are deployed at the access layer. These switches support 10GBaseT connections and provide VLAN capabilities, making them well-suited for connecting the six critical PCs per office, as seen in the logical LAN diagram.

To secure the network perimeter, two Palo Alto PA-5280 firewalls are used in a high-availability (HA) configuration. These next-generation firewalls deliver advanced threat protection and traffic inspection, securing both internal traffic and external communications.

Lastly, Cisco ASR 1001-X routers are deployed for WAN and ISP connectivity. Supporting 10GBASE-LR transceivers, they offer reliable long-distance fibre connectivity and enterprise-grade routing features necessary for connecting to optical WAN and external providers.

Table for Cable type and Connectors:

Symbol	Cable Type	Speed	Connector	Max Distance
	CAT6a Copper	10Gbps	RJ-45	100m
	10GBASE-SR	10Gbps	SFP+	400m
	40Gbps QSFP+ DAC	40Gbps	QSFP+	7m
	Single-Mode Fiber (10GBASE-LR)	10Gbps	SFP+	10km
	Single-Mode Fiber (100GBASE-SR4 80Gbps DWDM)	80Gbps	CFP2	125m

Overall, the chosen hardware supports a fully redundant, secure, and scalable network infrastructure, aligning directly with the logical and physical designs provided.

Chapter 2: Wide Area Network (WAN)

Logical WAN Diagram

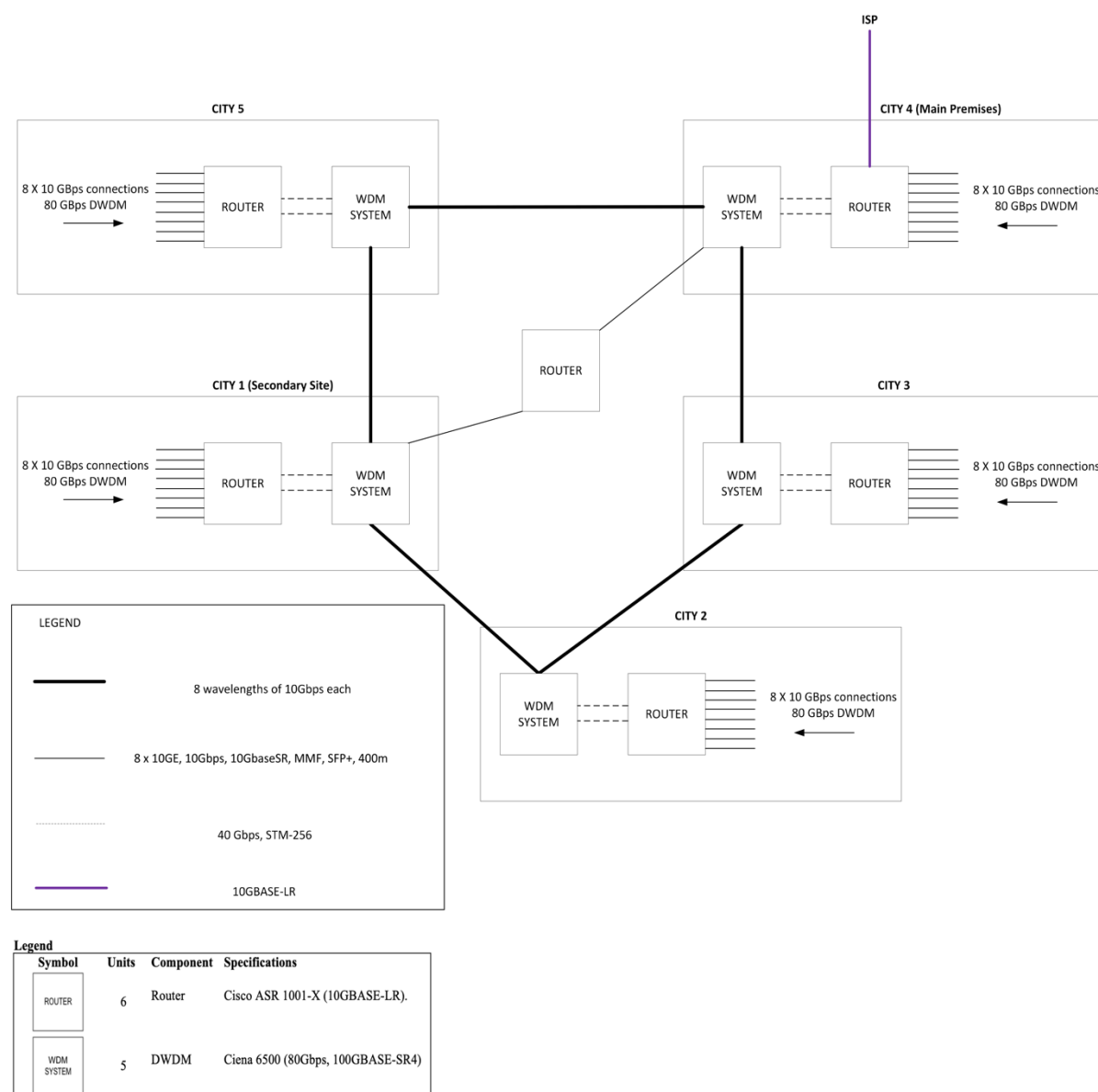


Figure 3: Logical WAN Diagram

2.1 Design Description and Justification

The WAN design establishes a high-performance, redundant optical link between the headquarters and the main premises of the food chain using Dense Wavelength Division Multiplexing (DWDM) technology. This architecture is essential for synchronising critical enterprise functions such as inventory management, supply chain operations, and ERP systems across geographically dispersed sites while ensuring maximum uptime and data continuity.

The network employs Ciena 6500 DWDM systems at both ends, functioning as carrier-grade optical multiplexers. These units aggregate multiple high-speed data streams over a single fibre pair. The transmission medium is Single-Mode Fibre (SMF) supporting 100GBASE-LR4, which is suitable for long-distance optical transport and provides a total capacity of 80 Gbps, achieved through eight 10Gbps wavelengths. This setup complies with the IEEE 802.3 standard and matches the fibre cabling configuration shown in the WAN diagram.

At each site, the DWDM systems are connected to Cisco ASR 1001-X routers using 10GBASE-LR SFP+ transceivers. These routers manage separate traffic streams—such as real-time sales data, cloud synchronization, and database replication—enhancing load balancing and segmenting network responsibilities. The use of multiple routers per site ensures that operational continuity is maintained even if a single device or link fails.

Redundancy is embedded in multiple layers of the WAN architecture:

1. Dual ASR routers per site provide fault tolerance through dynamic routing protocols, enabling rapid failover in case of device or link failure.
2. 10GBASE-LR optics facilitate long-distance, high-reliability fibre communication between sites, ensuring consistent data delivery over extended geographical areas.
3. The Ciena 6500 supports dual transponders and optical path protection ($\lambda 1/\lambda 2$), automatically rerouting traffic to alternative wavelengths in the event of physical or optical failures.

This DWDM-based topology ensures ultra-low latency, high availability, and robust fault tolerance—key requirements for mission-critical enterprise operations. The design strictly aligns with the WDM-to-router topology discussed in lectures and lab sessions and is visually represented in the logical WAN diagram, which clearly illustrates the redundant fibre paths and routing integration.

2.2 Network Hardware Selection and Evaluation

To support the WAN's stringent demands for speed, redundancy, and carrier-grade reliability, two key hardware components have been chosen: the Cisco ASR 1001-X routers and the Ciena 6500 DWDM platforms.

The Cisco ASR 1001-X is a compact, high-performance enterprise router designed for WAN edge environments. Each unit is equipped with 10GBASE-LR SFP+ interfaces and dual power supply units for hardware-level redundancy. These routers are highly scalable and support advanced features such as MPLS, BGP, and IPsec VPN, which are essential for securely extending

corporate services across wide geographic areas (Cisco, 2023). Their ability to handle concurrent routing, security, and traffic management tasks makes them ideal for interfacing directly with the optical backbone while maintaining flexible and secure traffic control.

Complementing the routers, the Ciena 6500 DWDM units function as the optical core of the WAN. These systems can multiplex up to 80 Gbps of traffic over single fibre pairs using 100GBASE-SR4 technology. The Ciena 6500 is widely deployed by telecommunications providers and large enterprises due to its modular design, support for intelligent wavelength switching, and integrated service awareness (Ciena, 2023). In this scenario, it serves as the inter-site transmission solution, futureproofing the organisation for further bandwidth growth without the need for additional physical cabling.

Together, these components enable a carrier-grade optical WAN architecture that is not only consistent with the logical design but also supports the strategic goals of the organization by ensuring performance, reliability, and long-term scalability.

Conclusion

This network design demonstrates how a structured hierarchical LAN and DWDM-based WAN can deliver high availability, performance, and scalability. By aligning with enterprise best practices and incorporating redundancy at multiple layers, the proposed architecture supports real-time business operations while reducing operational risk. The chosen hardware and topology not only fulfil the current network demands but also allow future growth, SDN integration, and digital transformation initiatives.

References

- Cisco Systems, 2023. *Cisco Nexus 9500 Series Switches*. [online] Available at: <https://www.cisco.com/c/en/us/products/switches/nexus-9500-platform-switches/index.html> [Accessed 8 Apr. 2025].
- Cisco Systems, 2023. *Cisco Catalyst 9300 Series Switches Data Sheet*. [online] Available at: <https://www.cisco.com/c/en/us/products/switches/catalyst-9300-series-switches/index.html> [Accessed 8 Apr. 2025].
- Cisco Systems, 2023. *Cisco Catalyst 2960-X Series Switches*. [online] Available at: <https://www.cisco.com/c/en/us/products/switches/catalyst-2960-x-series-switches/index.html> [Accessed 8 Apr. 2025].
- Cisco Systems, 2023. *Cisco ASR 1000 Series Aggregation Services Routers*. [online] Available at: <https://www.cisco.com/c/en/us/products/routers/asr-1000-series-aggregation-services-routers/index.html> [Accessed 8 Apr. 2025].
- Palo Alto Networks, 2023. *PA-5280 Data Sheet*. [online] Available at: <https://www.paloaltonetworks.com/resources/datasheets/pa-5200-series> [Accessed 8 Apr. 2025].
- Ciena, 2023. *Ciena 6500 Packet-Optical Platform*. [online] Available at: <https://www.ciena.com/products/6500/> [Accessed 8 Apr. 2025].
- IEEE, 2021. *IEEE Standard for Ethernet*. IEEE 802.3-2021. Institute of Electrical and Electronics Engineers. [online] Available at: https://standards.ieee.org/standard/802_3-2022.html [Accessed 8 Apr. 2025].